AdaFed: Adaptive Selective Aggregation for Heterogeneous Federated Learning in Autonomous Driving

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***Abstract*—We present a comprehensive empirical study of federated learning (FL) for seven heterogeneous autonomous driving models and propose AdaFed, an adaptive selective aggregation strategy. We introduce a tier-based partial backbone sharing framework that groups models by shared architectural components (ResNet-34/50 backbones, bird's-eye-view (BEV) encoders, and BEV-Lift modules) and evaluate six FL strategies: FedAvg, FedProx, SCAFFOLD, FedMD, FedDF, and AdaFed. Across all seven models, no FL strategy surpasses the per-model imitation-learning (IL) baseline; the practical question is therefore which strategy *minimises degradation*. AdaFed applies a tier-specific base strategy, blends the aggregated and local weights with an adaptive coefficient, and reverts any round-level update that worsens validation error. Experiments on CARLA 0.9.16 reveal that (1) partial backbone averaging stays closest to the IL baseline for ResNet-50 models (+0.05–0.13 m average displacement error (ADE)), (2) knowledge-distillation (KD) based approaches degrade performance, and (3) AdaFed yields the best FL result for the two models most exposed to negative transfer—VAD (2.05 m, -0.26 m vs. the best conventional FL strategy) and UniAD (1.52 m). These findings provide practical guidelines for heterogeneous FL deployment in autonomous driving.**

***Index Terms*—Federated Learning, Autonomous Driving, Heterogeneous Models, Adaptive Aggregation, Knowledge Distillation, CARLA**

I. Introduction

Autonomous driving systems employ diverse architectures, from camera-only planners (TCP, VAD, UniAD) to multi-sensor fusion models (TransFuser, InterFuser, LAV, ReasonNet), each with distinct backbone networks and planning modules. While individual training on proprietary datasets yields strong performance, the inability to share knowledge across organizational boundaries limits collective progress and data efficiency in an industry where driving data is expensive to collect.

Federated Learning (FL) enables privacy-preserving collaborative training without centralizing sensitive driving data. However, standard FL algorithms like FedAvg [1] assume model homogeneity—a fundamentally broken assumption when manufacturers deploy models with different parameter counts (21.8M–42.7M), backbone networks (ResNet-34 vs. ResNet-50), and task-specific heads.

The challenge is particularly acute because (a) shared backbone components (ResNet-34/50) interact differently with model-specific perception and planning heads, (b) naive aggregation of shared parameters can disrupt tightly coupled feature representations, causing negative transfer that *degrades* rather than improves performance, and (c) no single FL strategy works universally across diverse architectures. Indeed, in our study none of the six FL strategies recovers the per-model IL baseline; the realistic objective for privacy-preserving collaboration is thus to bound the unavoidable degradation rather than to expect an improvement. We address this with a *tier-based partial backbone sharing framework* and propose **AdaFed**, an adaptive selective aggregation method that assigns a base strategy per tier, blends aggregated and local weights with an adaptive coefficient, and reverts updates that demonstrably harm an individual model, thereby mitigating negative transfer.

Our contributions: (1) a tier-based sharing framework for heterogeneous FL in autonomous driving; (2) the first six-strategy comparison across seven diverse architectures; (3) AdaFed—achieving the best FL result for the models most vulnerable to negative transfer through tier-adaptive aggregation with adaptive blending and rollback; and (4) identification of critical architectural factors determining FL effectiveness in heterogeneous settings.

II. Related Work

Federated Learning. FedAvg [1] performs local stochastic gradient descent (SGD) followed by model averaging. FedProx [2] adds proximal regularization to constrain local updates near the global model, handling statistical heterogeneity. SCAFFOLD [3] uses server and client control variates for variance reduction, providing theoretical convergence guarantees. While effective for non-independent and identically distributed (non-IID) data, these methods fundamentally assume identical architectures across all clients.

Heterogeneous FL. FedMD [4] enables model-agnostic knowledge transfer via knowledge distillation (KD) on shared public data, computing consensus predictions for all models to learn from. FedDF [5] extends this with confidence-weighted ensemble distillation based on validation performance. Other approaches include FedGen (generative model-based transfer), FedProto (prototype sharing), and split learning variants. However, these have been evaluated primarily on classification tasks with simple architecture variations, not complex multi-modal driving systems. Adaptive FL methods explore selective participation [6], but tier-level selective aggregation for heterogeneous architectures remains unexplored.

Autonomous Driving Models. TCP [7] uses trajectory-conditioned planning with ResNet-34. TransFuser [8] fuses camera and LiDAR through transformer attention. InterFuser [9] employs ResNet-50 with interpretable sensor fusion. LAV [10] learns from all vehicles. ReasonNet [11] incorporates temporal reasoning. VAD [12] introduces vectorized scene representations, and UniAD [13] proposes a unified planning framework. These share ResNet backbone families but differ in downstream modules, presenting a natural testbed for heterogeneous FL.

III. Method

A. Tier-Based Partial Sharing

We consider K=7 autonomous driving models with distinct architectures. Despite heterogeneity, we identify four sharing tiers (Fig. 1): **Tier 1** (`resnet34`): TCP, TransFuser, LAV—3 models, 21.3M params, 216 keys. **Tier 2** (`resnet50`): InterFuser, ReasonNet, VAD, UniAD—4 models, 23.6M params, 318 keys. **Tier 3** (`bev_encoder`): TransFuser, InterFuser, LAV, ReasonNet—120 keys. **Tier 4** (`bev_lift`): VAD, UniAD—5.9M params, 30 keys. Only tier-matched parameters are aggregated; model-specific heads remain locally updated.

B. Baseline FL Strategies

FedAvg [1]: Tier-wise weight averaging. **FedProx** [2]: + proximal term (μ=0.01). **SCAFFOLD** [3]: Control variate drift correction. **FedMD** [4]: Consensus KD on public data. **FedDF** [5]: Confidence-weighted ensemble KD.

C. AdaFed: Adaptive Selective Aggregation

Existing strategies apply aggregated weights blindly, which can harm models whose features are disrupted by averaging. AdaFed combines three mechanisms (Algorithm 1). Throughout, θ_k denotes the full parameter set of model k, while $w^{k,t}$ denotes only the subset of parameters that model k shares within tier t; aggregation and blending act on the tier-level w, whereas snapshots and rollback act on the full θ.

Tier-specific base strategy. Rather than wrapping a single algorithm, AdaFed assigns each tier the base strategy that our empirical comparison (Sec. V-B) found most effective for that tier: SCAFFOLD for `resnet34`, FedAvg for `resnet50` and `bev_encoder`, and FedProx for `bev_lift`.

Soft blending. After local training, model k holds local shared weights $w_{\text{local}}^{k,t}$ for each tier t, and the server computes a tier average w_{avg}^{t} over the models in that tier. Rather than overwriting the model with w_{avg}^{t}, AdaFed blends the two with an adaptive coefficient $\alpha_k^t \in [\alpha_{\min}, \alpha_{\max}]$:

$$w_{\text{new}}^{k,t} = \alpha_k^t \, w_{\text{avg}}^t + (1-\alpha_k^t)\, w_{\text{local}}^{k,t}. \tag{1}$$

Algorithm 1 AdaFed: Adaptive Selective Aggregation

Require: Models $\{M_k\}_{k=1}^K$, tiers $\{T_t\}$, rounds R
Init: assign base strategy per tier; $\alpha_k^t \leftarrow \alpha_{\text{init}}$
Init: establish baseline $\text{ADE}_k^{\text{cur}}$ from IL weights
for round $r=1$ to R **do**
  Snapshot full state θ_k of every model (for rollback)
  for each model k **do**
    Local training of M_k (with tier-specific base strategy: SCAFFOLD correction / FedProx term as applicable)
  end for
  for each tier t **do**
    Compute tier average w_{avg}^t
    for each model $k \in T_t$ **do**
      $w_{\text{new}}^{k,t} \leftarrow \alpha_k^t w_{\text{avg}}^t + (1-\alpha_k^t) w_{\text{local}}^{k,t}$
      Inject $w_{\text{new}}^{k,t}$ into M_k
    end for
  end for
  for each model k **do**
    Evaluate $\text{ADE}_k^{\text{new}}$ on the validation set
    if $\text{ADE}_k^{\text{new}} \leq \text{ADE}_k^{\text{cur}}$ **then**
      Accept: $\text{ADE}_k^{\text{cur}} \leftarrow \text{ADE}_k^{\text{new}}$; $\alpha_k^t \mathrel{+}= \delta^+$
    else
      Rollback: restore snapshot θ_k; $\alpha_k^t \mathrel{-}= \delta^-$
    end if
  end for
end for

A larger α trusts the shared aggregate more; a smaller α keeps the model closer to its own weights. The coefficient is maintained per model and per tier, so different tiers of the same model can trust aggregation to different degrees.

Validation-based rollback and α adaptation. After blending, each model is validated. If its ADE does not worsen relative to the model's current best, the update is accepted and α is increased by δ^+; if the ADE worsens, the entire round is rolled back to the pre-round snapshot and α is decreased by δ^-. Rollback therefore reverts a round that has already been trained and aggregated, rather than skipping computation; its purpose is to prevent a harmful update from persisting, not to save time.

Through rollback, a model whose blended update would degrade it returns to its pre-round state, so its ADE stays constant across that round; through α adaptation, models repeatedly helped by aggregation come to trust it more, while models repeatedly harmed retreat toward their own weights.

IV. Experimental Setup

A. Models and Data

We evaluate seven models (Table I) on driving data collected from the CARLA 0.9.16 simulator: 23,580 training and 5,895 validation samples across 15 episodes. The heterogeneity studied here is *architectural*: all clients share the same driving data, and each client holds a model with a

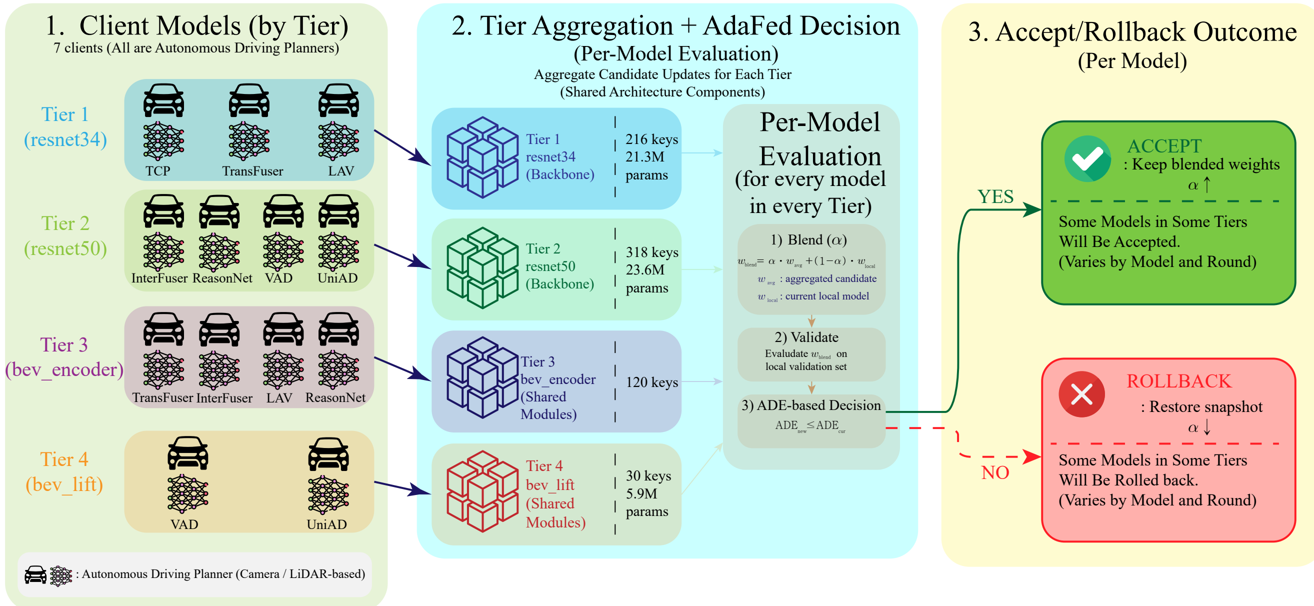

Fig. 1. AdaFed framework overview. Seven models are grouped into four architecture-sharing tiers (`resnet34`, `resnet50`, `bev_encoder`, `bev_lift`), each with a tier-specific base strategy. Aggregated and local weights are blended per model; the blended round is then validated and either accepted or rolled back to the pre-round snapshot.

TABLE I
MODEL ARCHITECTURE CONFIGURATION

Model	Params	Backbone	Shared Tiers
TCP	21.8M	ResNet-34	resnet34
TransFuser	37.0M	ResNet-34	resnet34, bev_enc
InterFuser	42.7M	ResNet-50	resnet50, bev_enc
LAV	35.5M	ResNet-34	resnet34, bev_enc
ReasonNet	40.2M	ResNet-50	resnet50, bev_enc
VAD	36.4M	ResNet-50	resnet50, bev_lift
UniAD	35.1M	ResNet-50	resnet50, bev_lift

TABLE II
IMITATION LEARNING BASELINE RESULTS

Model	Val Loss	ADE (m)	Epoch	Time
TCP	1.077	1.70	6/16	34.4m
TransFuser	1.028	1.72	9/19	69.1m
InterFuser	0.768	**1.27**	12/22	210.6m
LAV	1.219	2.09	5/15	36.6m
ReasonNet	0.783	**1.30**	11/21	194.6m
VAD	0.911	1.55	9/19	73.4m
UniAD	0.858	1.44	10/20	83.3m

different architecture. This is a controlled setting that isolates the effect of architectural heterogeneity from that of data heterogeneity; combining AdaFed with non-IID client data partitions is left to future work. For the KD strategies (FedMD, FedDF), a separate public subset (episodes reserved for this purpose, 19,650 samples) is used solely to compute consensus predictions.

B. Training Protocol and Environment

All experiments are conducted on a system with Intel Core i9-14900K CPU, 128GB RAM, and NVIDIA RTX 5090 GPU, using PyTorch 2.6.0 with CUDA 12.8.

IL Baseline: imitation learning (IL) with Adam (lr=10^{-4}, weight decay=10^{-4}), waypoint L1 loss, early stopping (patience 10). **FL Training:** starting from IL weights, 10 rounds, 2 local epochs/round, lr=5×10^{-5}. All six strategies are evaluated under identical conditions. We report Average Displacement Error (ADE) in meters; AdaFed's acceptance test uses the same validation set used for reporting, and all results are from a single run per strategy. Sec. VI-F discusses both points.

V. RESULTS

A. IL Baseline

Table II presents the IL baseline results. ResNet-50 models consistently outperform ResNet-34: InterFuser achieves the best ADE (1.27m), followed by ReasonNet (1.30m), UniAD (1.44m), and VAD (1.55m). Among ResNet-34 models, TCP (1.70m) and TransFuser (1.72m) perform similarly, while LAV lags at 2.09m. The results suggest that backbone depth is a stronger performance predictor than BEV encoding alone: LAV uses `bev_encoder` but with ResNet-34, performing worse than VAD which uses `bev_lift` with ResNet-50.

B. FL Strategy Comparison

Table III presents the comprehensive comparison. No conventional FL strategy improves upon the IL baselines; the degradation, however, varies dramatically across strategies and architectures.

FedAvg stays closest to the IL baseline for ResNet-50 models: InterFuser (+0.06m), ReasonNet (+0.05m), UniAD

(+0.13m). The 4-model `resnet50` tier enables robust averaging. ResNet-34 models suffer more (+0.54–1.47m).

FedProx is the best conventional strategy for VAD (2.31m vs. FedAvg's 2.42m), suggesting `bev_lift` benefits from proximal constraints.

SCAFFOLD is the strongest strategy for all ResNet-34 models (TCP: 2.39m, TransFuser: 2.30m, LAV: 2.46m) but catastrophically fails for VAD (5.55m). Control variates handle the 3-model tier well but destabilize `bev_lift`.

FedMD and FedDF degrade performance the most. Best ADEs occur at round 1, with subsequent rounds worsening—consensus from diverse architectures introduces conflicting gradients rather than useful knowledge.

These per-tier observations directly motivate AdaFed's tier-specific base-strategy assignment (Sec. III-C).

C. AdaFed Analysis

AdaFed gives the best FL result for the two models most vulnerable to negative transfer:

VAD: 2.05m (-0.26m vs. FedProx's 2.31m, the best conventional strategy; still $+0.50$m vs. the 1.55m IL baseline). VAD's `bev_lift` tier (shared only with UniAD) makes it susceptible to blind aggregation. As Fig. 2 shows, the convergence is a *staircase*: ADE drops at rounds whose blended update is accepted and stays flat at rounds that are rolled back (3.92m at R1, then 3.41, 3.32, 2.71, 2.45, and 2.05m at R10). This is the closest any FL strategy comes to VAD's IL baseline.

UniAD: 1.52m (-0.05m vs. FedAvg's 1.57m; $+0.08$m vs. the 1.44m IL baseline). UniAD follows the same staircase pattern (2.66m at R1 down to 1.52m at R9; Fig. 2). Adaptive blending lets UniAD increasingly trust `resnet50` tier averaging while rollback protects its unified planning module from disruptive `bev_lift` updates.

Rollback patterns. TransFuser's ADE remains constant at 3.04m across all 10 rounds, and LAV at 3.06m (Fig. 2, flat curves): every round is rolled back, so each model stays at its post-warm-up state because tier averaging never improves it. TCP, in contrast, improves from 3.22m at R1 to 2.54m, showing that some ResNet-34 models do benefit from accepted rounds. InterFuser reaches 1.36m (close to FedAvg's 1.33m), showing AdaFed preserves performance for well-served models.

D. Convergence Behavior

FedAvg/FedProx converge gradually (best at R8–10). SCAFFOLD converges faster (R1–5) but oscillates for some models. KD strategies diverge after R1 with monotonically increasing ADE—FedMD's VAD reaches 19.4m at R1. AdaFed shows a distinctive *staircase* convergence (Fig. 2): models improve in discrete accepted steps, and each flat segment marks a rolled-back round, making the trajectory directly interpretable.

VI. Discussion

A. Architecture-Dependent Strategy Selection

Table IV reveals three distinct strategy regimes. ResNet-34 models favor SCAFFOLD for variance reduction in the 3-model tier. ResNet-50 models with `bev_encoder` (InterFuser, ReasonNet) favor FedAvg in the 4-model tier. `bev_lift` models (VAD, UniAD) benefit from AdaFed's adaptive blending and rollback. This regime structure is exactly what AdaFed's tier-specific base-strategy assignment exploits.

B. When Does AdaFed Help?

AdaFed excels for models in *small or mismatched tiers*. The `bev_lift` tier (2 models) is where blind averaging most disrupts features, and AdaFed's rollback provides an effective safety net. For well-matched large tiers (InterFuser, ReasonNet in the 4-model `resnet50` tier), FedAvg already performs well and AdaFed yields comparable results (1.36m vs. 1.33m). For ResNet-34 models, AdaFed's rollback simply reverts to the post-warm-up state every round, so it neither helps nor harms beyond that point—it is complementary to, not a replacement for, SCAFFOLD's variance correction.

C. The Heterogeneity-Performance Trade-off

Large, well-matched tiers tolerate simple averaging (+0.05–0.13m). Medium tiers need variance correction (SCAFFOLD). Small tiers need selective protection (AdaFed). VAD's poor performance across conventional strategies despite sharing a ResNet-50 backbone reveals that backbone similarity alone is insufficient—downstream module compatibility (vectorized scene representation vs. standard BEV) is equally critical. Future frameworks should assess full end-to-end architectural compatibility, not just shared sub-component identification.

D. Limitations of KD in Heterogeneous Settings

FedMD and FedDF's consistent underperformance reveals fundamental challenges with model-agnostic KD for complex driving models. When camera-only models (TCP, VAD) and multi-sensor fusion models (TransFuser, InterFuser) generate predictions on the same input, the averaged consensus fails to capture any individual model's strengths, producing conflicting gradients that actively harm performance. Future work should explore modality-aware or architecture-grouped distillation strategies.

E. Practical Implications

Because the optimal strategy depends on tier composition, a single global FL algorithm is suboptimal for heterogeneous driving fleets. AdaFed operationalises a tier-aware policy—SCAFFOLD-based aggregation for ResNet-34 tiers, FedAvg-based for large ResNet-50 tiers, FedProx-based for the small `bev_lift` tier—and adds rollback so that a tier whose update turns out harmful is reverted rather than propagated. KD-only approaches should be avoided.

F. Limitations and Threats to Validity

Four limitations should be noted. First, all results come from the CARLA simulator; simulated driving does not capture the sensor noise, label imperfection, and fleet diversity of real deployments. Validation on real-world multi-fleet driving datasets is needed to confirm practical applicability and is our

TABLE III

COMPREHENSIVE COMPARISON: IL BASELINE AND SIX FL STRATEGIES (ADE IN METERS). R = BEST ROUND. Δ = BEST FL − IL (ALL POSITIVE: EVERY STRATEGY DEGRADES VS. IL). **BOLD** = BEST FL PER MODEL. GREEN = ADAFED GIVES THE BEST FL RESULT.

Model	IL	FedAvg	R	FedProx	R	SCAFF.	R	FedMD	R	FedDF	R	AdaFed	R	Best	Δ
TCP	1.70	3.17	10	3.04	10	**2.39**	4	4.19	2	3.26	1	2.54	8	SCAFF.	+0.69
TransFuser	1.72	2.54	5	2.57	3	**2.30**	5	2.68	1	2.32	1	3.04	9	SCAFF.	+0.58
InterFuser	1.27	**1.33**	3	1.34	9	2.25	3	1.69	1	2.26	1	1.36	10	FedAvg	+0.06
LAV	2.09	2.63	8	2.67	3	**2.46**	9	3.03	1	3.04	2	3.06	1	SCAFF.	+0.37
ReasonNet	1.30	**1.35**	9	1.35	2	2.06	1	3.15	1	2.77	1	1.46	7	FedAvg	+0.05
VAD	1.55	2.42	10	2.31	10	5.55	4	5.75	3	4.41	2	**2.05**	10	AdaFed	+0.50
UniAD	1.44	1.57	10	1.64	10	2.25	9	2.80	1	2.34	1	**1.52**	9	AdaFed	+0.08

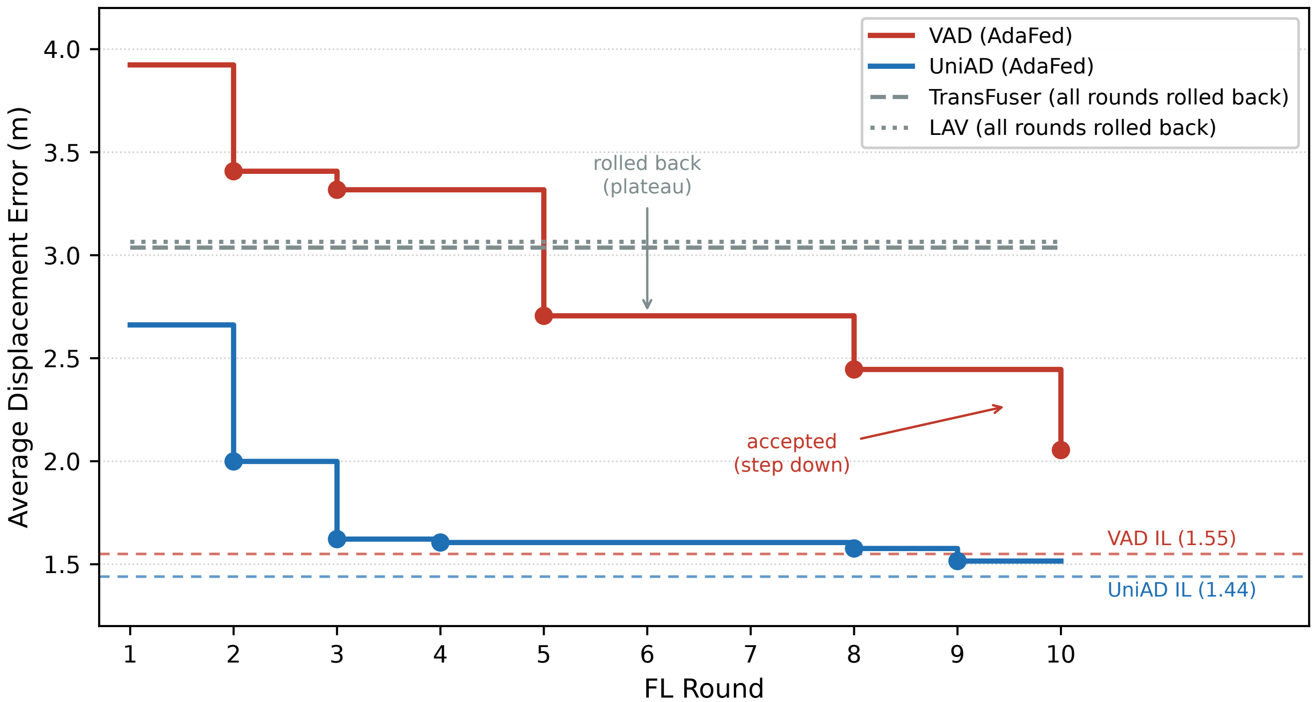

Fig. 2. Round-by-round ADE under AdaFed. VAD and UniAD improve in discrete accepted steps (markers) with plateaus at rolled-back rounds, while TransFuser and LAV remain flat because every round is rolled back. Dashed lines mark the IL baselines, which no FL strategy reaches.

TABLE IV

BEST FL STRATEGY PER MODEL

Model	Backbone	IL	Best FL	Strategy	Δ
TCP	ResNet-34	1.70	2.39	SCAFFOLD	+0.69
TransFuser	ResNet-34	1.72	2.30	SCAFFOLD	+0.58
InterFuser	ResNet-50	1.27	1.33	FedAvg	+0.06
LAV	ResNet-34	2.09	2.46	SCAFFOLD	+0.37
ReasonNet	ResNet-50	1.30	1.35	FedAvg	+0.05
VAD	ResNet-50	1.55	**2.05**	**AdaFed**	+0.50
UniAD	ResNet-50	1.44	**1.52**	**AdaFed**	+0.08

main planned follow-up. Second, the studied heterogeneity is architectural only: all clients share the same training data, so the results do not yet speak to combined architecture- and data-heterogeneous settings. Third, AdaFed's acceptance test is computed on the same validation set used for reporting; an independent held-out selection set would remove any optimistic bias and is a straightforward improvement. Fourth, each strategy is reported from a single run; because AdaFed's acceptance test depends on small ADE differences, multi-seed experiments with reported variance—and a small acceptance margin to absorb seed noise—would strengthen the staircase findings.

VII. CONCLUSION

We present AdaFed, an adaptive selective aggregation strategy for heterogeneous FL in autonomous driving. Using a tier-based partial backbone sharing framework across seven diverse models and six FL strategies, we show that the optimal FL approach depends critically on tier composition and architectural compatibility.

Key findings: (1) no FL strategy recovers the per-model IL baseline, so the realistic goal is to minimise degradation; (2) FedAvg with partial backbone sharing stays closest to baseline for ResNet-50 models (+0.05–0.13m ADE); (3) SCAFFOLD is the best choice for ResNet-34 models through variance reduction; (4) KD-based approaches (FedMD, FedDF) are insufficient for heterogeneous driving FL, with consensus distillation actively degrading performance; and (5) AdaFed, by assigning a tier-specific base strategy and adding adaptive blending with validation-based rollback, gives the best FL result for the most vulnerable models (VAD: 2.05m, UniAD: 1.52m).

Our analysis suggests that practical deployments should employ tier-aware strategies rather than a single global approach. Future directions include adaptive tier reassignment based on training dynamics, an independent selection set for the acceptance test, multi-seed validation, and evaluation on real-world multi-fleet driving datasets.

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